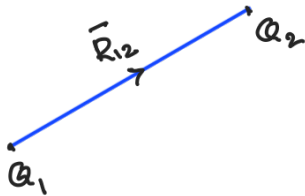


Electric field:

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0} \frac{1}{|R_{12}|^2} \hat{a}_{12}$$

We can say $\vec{F}_{12} \propto Q_2$

$$\vec{F}_{12}' = 4\vec{F}_{12} ; \text{ if } Q_2 \rightarrow 4Q_2 \dots$$

So an useful definition is;

Force / Coulomb of Q_2 = Electric field.

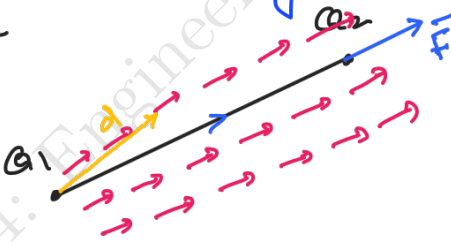
ie;
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{|R_{12}|^2} \hat{a}_{12} \text{ N/C}$$
 After introducing electro static potential we might use v/m.

$$\Rightarrow E = \frac{\vec{F}}{Q_2}$$

If we already know vector $\vec{F}_{12}$ (Force), why we need new vector $\vec{E}$ (Electric field)?

* It has something to do with how we interpret Coulomb's law.

This stress picture is useful to explain, quantum mechanics, relativity etc



- * The presence of the charge Q_1 polarized the material
- * ie in the presence of charge the material got stressed!
- * Any time we place another charge Q_2 in this stressed material, it receives a kick. This kick is called force!



$$\vec{E} = \frac{Q_1}{4\pi\epsilon_0} \frac{1}{d^2} \hat{d}$$

↳ But here
no second charge
location mentioned

The previous Eqn

$$\vec{E} = \frac{Q_1}{4\pi\epsilon_0} \frac{1}{|R_{12}|^2} \hat{R}_{12} \rightarrow \text{was electric field experienced at position of } Q_2 \text{ due to } Q_1.$$

* It is similar to saying
all the points around Q_1 , there exist
potential force waiting to be happened

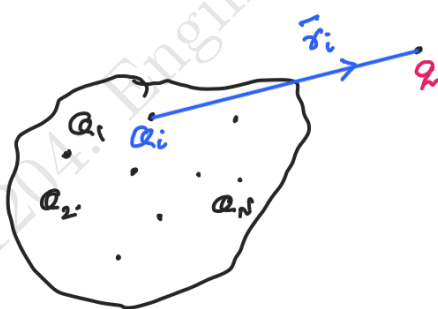
$$\vec{E}(\vec{d}) = \frac{Q_1}{4\pi\epsilon_0} \frac{1}{d^2} \hat{d}$$

$$= d_1\hat{i} + d_2\hat{j} + d_3\hat{k}$$

← It is now a function of independent
variable $\vec{d}$, which is also a vector.
ie; Electric field is a vector field.

Basically it is another form of Coulomb's law, except
that there is no second charge; and it is applied
to any arbitrary point in space.

Superposition principle:



$$\vec{E} = \frac{Q}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{r_i^2} \hat{r}_i$$

$\vec{r}_i \rightarrow$ distance vector between charge
 Q_i and Q

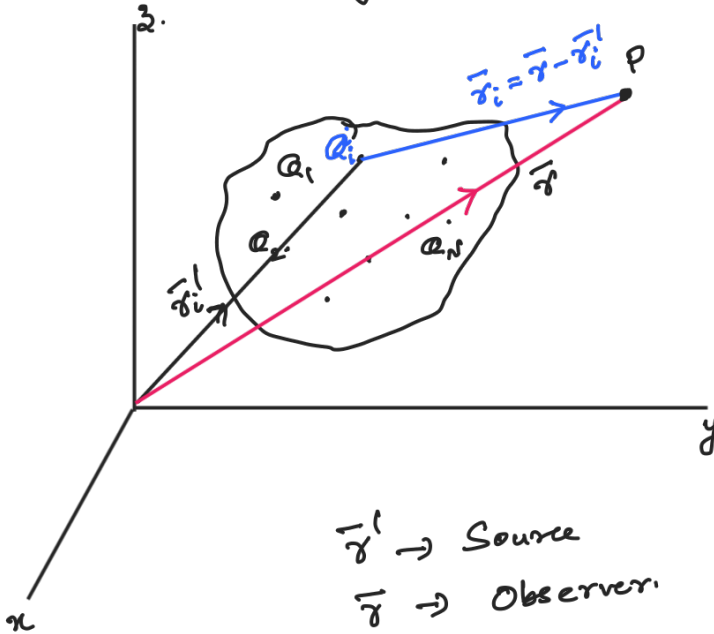
Similarly ; by superposition principle, $\vec{E}$ at any point $\vec{r}$

Similar to keeping a test charge of 1C at $\vec{r}$, and measuring the force.

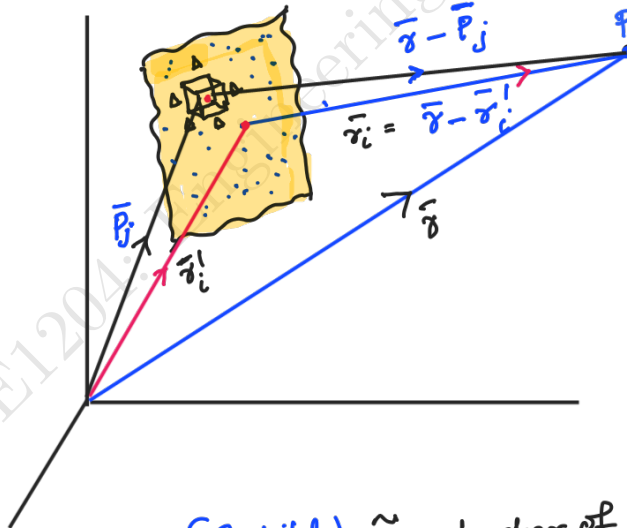
$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{|\vec{r} - \vec{r}_i'|^2} \frac{(\vec{r} - \vec{r}_i')}{|\vec{r} - \vec{r}_i'|}$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{|\vec{r} - \vec{r}_i'|^3} (\vec{r} - \vec{r}_i')$$

independent variable.



Electric field due to a cloud of charges distributed over a region
'N' charges where $N \gg \sim 10^{23}$

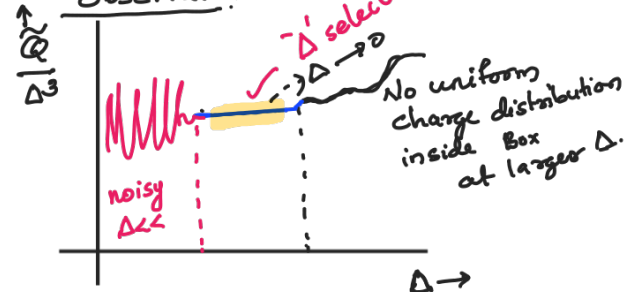


Box of size Δ ; then

Δ	$\tilde{Q}$	$\sim Q/\Delta^3$
1cm	...	...
1mm	...	...
1μm	...	...

Number of charges decreases

Experimental observation



$(Q \text{ tilda}) \rightarrow \tilde{Q} = \text{Number of charges inside a box of size } \Delta$

→ With reliable Δ sized boxes we can construct entire volume with boxes. $\hookrightarrow \frac{\tilde{Q}}{\Delta^3}$ Constant

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{|\vec{r} - \vec{r}_i'|^3} (\vec{r} - \vec{r}_i')$$

independent variable.

Here we count the number of particles/charges.

$$\tilde{Q}_j = \frac{\tilde{Q}_j}{\Delta^3} \times \Delta^3$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^M \frac{\tilde{Q}_j \cdot \Delta^3}{|\vec{r} - \vec{r}_j|^3} (\vec{r} - \vec{r}_j)$$

$\frac{\tilde{Q}_j}{\Delta^3}$: Average charge / unit volume
 $\Rightarrow$ Charge density (ρ)

As $\Delta \rightarrow 0$

(only within the limit where $\frac{\tilde{Q}}{\Delta^3}$ is constant.)

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dV$$

$\Rightarrow$ most used equation in electrostatics.

This sum now will become the following;

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^M \frac{\tilde{Q}_j}{|\vec{r} - \vec{r}_j|^3} (\vec{r} - \vec{r}_j)$$

summing over boxes.

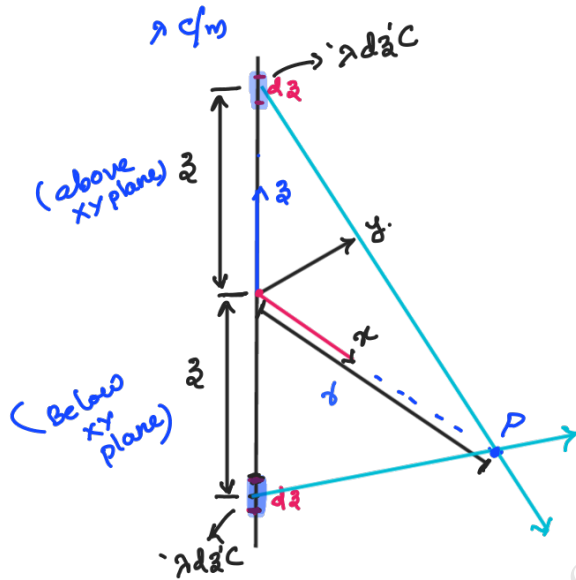
$\Rightarrow$ Integration over the volume

Example 1: Line charge of λ C/m ; find $\vec{E}$ at point P.

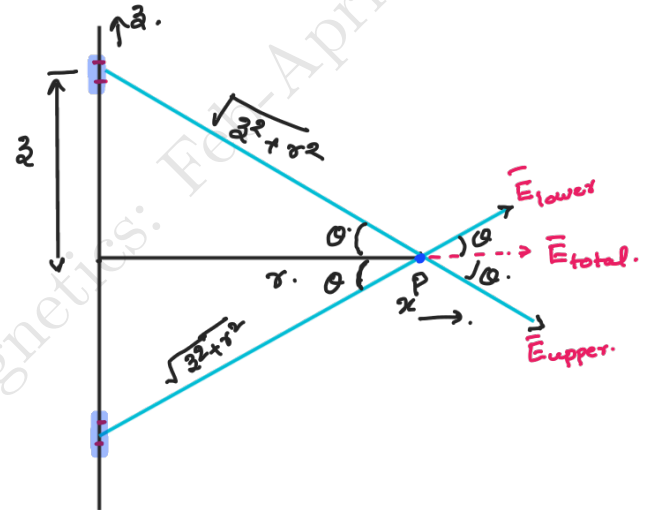
(Text book examples)

Charge/unit length, $= \lambda$ C

Solution: Let us put line charge along z axis of co-ordinate dm as shown below;



$\equiv$



$$|\vec{E}_{\text{lower}}| = |\vec{E}_{\text{upper}}| = \frac{\lambda dz}{4\pi\epsilon_0} \cdot \frac{1}{(z^2 + x^2)}$$

Total Electric Field due to dz (upper + lower) $|\vec{E}_{\text{total}}| = |\vec{E}_{\text{lower}}| \cos\theta + |\vec{E}_{\text{upper}}| \cos\theta$

$$d\vec{E}_{\text{total}} = \frac{2\lambda dz}{z^2 + x^2} \cos\theta \hat{i}$$

$$\vec{E} = \hat{i} \int_0^\infty \frac{\lambda dz}{4\pi\epsilon_0} \frac{2\cos\theta}{z^2 + x^2} \quad [z^2 \rightarrow x^2 \text{ as } z \text{ in the } x \text{ coordinate}]$$

$$\cos\theta = \frac{x}{\sqrt{z^2 + x^2}}$$

$$\vec{E} = \hat{i} \int_0^{\infty} \frac{\lambda dz}{4\pi\epsilon_0} \frac{2x}{(z^2+x^2)^{3/2}}$$

$$E_x = \frac{2\lambda x}{4\pi\epsilon_0} \int_0^{\infty} \frac{dz}{(z^2+x^2)^{3/2}}$$

$$= \frac{2\lambda x}{4\pi\epsilon_0} \int_0^{\pi/2} \frac{x \sec^2\phi d\phi}{x^3 \sec^3\phi}$$

$$= \frac{2\lambda}{4\pi\epsilon_0} \cdot \frac{1}{x} \int_0^{\pi/2} \underbrace{\cos\phi d\phi}_1$$

$$z = x \tan\phi.$$

$$dz = x \sec^2\phi d\phi$$

$$(z^2+x^2)^{3/2} = (x^2 \tan^2\phi + x^2)^{3/2}$$

$$= x^3 (1+\tan^2\phi)^{3/2}$$

$$= x^3 \sec^3\phi.$$

$$z \rightarrow 0 \quad \phi \rightarrow 0.$$

$$z \rightarrow \infty \quad \phi \rightarrow \pi/2.$$

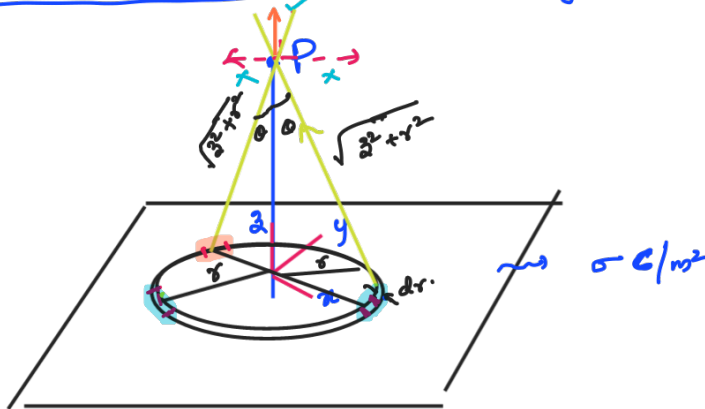
$$\vec{E} = \hat{i} \frac{\lambda}{2\pi\epsilon_0} \cdot \frac{1}{x}.$$

$$\boxed{E(\vec{r}) = \frac{\lambda}{2\pi\epsilon_0} \frac{1}{r} \hat{r}} \rightarrow \text{In Radial direction.}$$

* Due to line charge $\vec{E} \propto \frac{1}{r}.$

* Why due to line charge $\vec{E} \propto \frac{1}{r}$ but not $\propto \frac{1}{r^2}$ like point charge!
Intuitive explanation?



Electric Field due to a plane charge.

Solution: Consider a ring centered around the xy-location of the point
Let radius of the ring be r and thickness be dr :

The amount of charge in the ring
 $dQ = \sigma (2\pi r dr)$

* All the points on the ring is equidistant from the point 'P'

* For every point on the ring, there exist an equal & opposite electric field. $\Rightarrow$ Only the vertical component survive

$\Rightarrow$

$$dE_z = \frac{2\pi r \sigma dr}{4\pi \epsilon_0} \frac{1}{(z^2 + r^2)^{3/2}} \underbrace{2 \cos \theta}_{\text{From equal \& opposite components on the ring.}}$$

$$\cos \theta = \frac{z}{\sqrt{z^2 + r^2}}$$



$$\Rightarrow dE_z = \frac{2\pi r \sigma dr}{4\pi \epsilon_0} \frac{2z}{(z^2 + r^2)^{3/2}}$$

$$E_z = \frac{2\pi \sigma}{4\pi \epsilon_0} z \int_0^\infty \frac{r dr}{(z^2 + r^2)^{3/2}}$$

$$= \frac{2\pi \sigma}{4\pi \epsilon_0} z \int_{z^2}^\infty \frac{du/2}{u^{3/2}}$$

$$= \frac{2\pi \sigma}{4\pi \epsilon_0} z \left[\frac{1}{1/2} \times \frac{u^{-1/2}}{-1/2} \right]_{z^2}^\infty$$

$$= \frac{2\pi \sigma}{4\pi \epsilon_0} z \cdot \frac{1}{z}$$

$$= \frac{2\pi \sigma}{4\pi \epsilon_0} = \frac{\sigma}{2\epsilon_0} = \underline{\underline{\text{Constant}!!}}$$

(Removed factor $\frac{1}{2}$ as $r=0 \rightarrow \infty$ cover entire plane).

$$z^2 + r^2 = u.$$

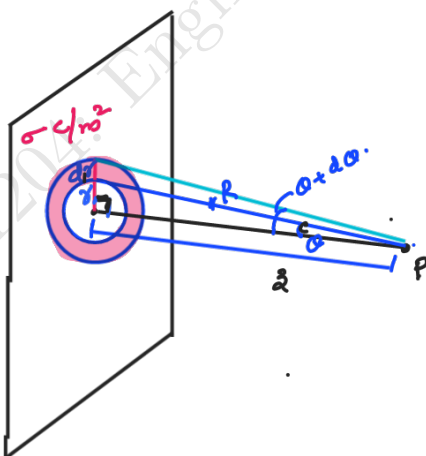
$$du = 2r dr.$$

$$\begin{matrix} r \rightarrow 0 & u \rightarrow z^2 \\ r \rightarrow \infty & u \rightarrow \infty \end{matrix}$$

$$\frac{u^{-3/2+1}}{-3/2+1}$$

$$= \frac{u^{-1/2}}{-1/2}$$

Intuitive explanation:



Understanding dQ (charge in the ring)

* When extend an angle $d\theta$ from 'P' to the plane, it intercept a ring of radius ' $r = z \tan \theta$ '.

* Between θ & $\theta + d\theta$, it forms a thin ring with area

$$dA \approx (2\pi r) dr$$

$$\approx 2\pi (z \tan \theta) d(z \tan \theta)$$

$$\approx (2\pi \tan \theta \sec^2 \theta \cdot z) z^2$$

Thus $dQ \propto \sigma dA$

$$\propto \sigma z^2 \tan \alpha \sec^2 \alpha d\alpha$$

* So as ' z ' increases dQ increases. (as ring get larger & larger)

2. Understand $\frac{1}{R^2}$ (inverse square effect)

Distance from any charge element in the ring to 'P' is;

$$R = \frac{z}{\cos \alpha} \Rightarrow \frac{1}{R^2} = \frac{\cos^2 \alpha}{z^2}$$

* It shows as ' z ' increases, the field contribution from each charge decreases.

3. How they cancel out?

a) The charge element increases $\Rightarrow dQ \propto z^2$

b) The field contribution / charge decreases as $\frac{1}{R^2} \propto \frac{1}{z^2}$.

c) Together, the total contributions to the field from each ring;

$$dE \propto \frac{dQ}{R^2} \propto \frac{z^2 \tan \alpha \sec^2 \alpha d\alpha \cos^2 \alpha}{z^2} \propto \tan \alpha d\alpha$$

* The ' z ' term cancel out, meaning the field does not depend on ' z '.

* Only angular terms remain, and if summed over all ' α ' the result must be constant



Key difference from infinite plane case to finite plane charge:

- * For infinite plane, as ' z ' increases, larger rings keep contributing more charge.
- * For finite plane, there is a limit - beyond a certain ' z ', there is no more rings to add charge from as the plane ends.
 $\Rightarrow dQ$ does not increase indefinitely, so it does not fully compensate for decrease due to $\frac{1}{r^2}$.

Example: Try to find electric field due to a disk of radius ' a ' and surface charge density $\sigma \text{ C/m}^2$.

$$\text{Ans: } \vec{E} = \hat{z} \frac{\sigma z}{2\epsilon_0} \left(\frac{1}{z} - \frac{1}{\sqrt{z^2 + a^2}} \right)$$